Abhinand
Department of Mechanical Engineering
Indian Institute of Technology Madras

IIT Madras

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*All results reproducible from committed artifacts*

# Outline


1. Introduction
2. Related Work
3. Digital Twin Physics
4. Environment Design
5. Reinforcement Learning
6. Results
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8. Software Engineering
9. Discussion
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# Problem: Dynamic Cutting Parameter Optimization

CNC face milling: rotating cutter removes material in multiple passes.

**Parameters to optimize:**

- Spindle speed $n$ [RPM]
- Feed per revolution $f_r$ [mm/r]
- Axial depth of cut $a_p$ [mm]

**Four competing objectives:**

- **Productivity:** Maximize $\mathrm{MRR} = v_c \cdot a_p \cdot f_r$
- **Energy:** Minimize total electrical energy/part
- **Quality:** $R_a \leq 1.6$ $\mu$m (finished surface)
- **Tool Life:** $V_B \leq 0.3$ mm, avoid breakage

Traditional fixed parameters (handbooks, static NSGA-II) cannot adapt to tool wear, material variation, or geometry changes.

# Proposed Approach

1. **Digital Twin (Physics)**
   - Speed-dependent cutting energy $k_c(v_c) = k_{c0}(v_c/v_{\text{ref}})^m$
   - Extended Taylor tool wear
   - Empirical roughness regression
   - Stochastic breakage
2. **DRL Agent (PPO)**
   - One Gymnasium environment
   - 11-D observation, context always present
   - Energy-aware reward
   - Conditioned policy ($w_2$ + material context)
3. **Experimental Rigor**
   - 100-scenario Monte Carlo + 95% CIs
   - Ablation: 7 weights $\times$ 3 seeds
   - 5 classical baselines + receding-horizon
   - Robustness: 2 regimes against the controller
   - Regime map: two sweeps, the noise and the material model error
   - Batch: one tool, ten parts
   - Transfer experiment
   - Full reproducibility
4. **Interactive Workstation**
   - 2D stock depletion, micro-topography
   - G-code generation, multi-controller comparison

# Positioning Against Reference Papers

| **Capability** | Mo et al. (2025) *J. Manuf. Process 156* | Wang et al. (2025a) *J. Manuf. Syst. 78* | **This Work** |
|---|---|---|---|
| Dynamic optimization | ✓ | ✓ | ✓ |
| Tool wear in the state | ✓ | ✗ | ✓ |
| Roughness constraint | ✓ | ✗ | ✓ |
| Energy as an objective | ✗ | ✗ | ✓ |
| Speed-dependent $k_c$ | ✗ | ✗ | ✓ |
| Domain randomization | ✗ | ✗ | ✓ |
| Conditioned policy | ✗ | ✗ | ✓ |
| Transfer experiment | ✗ | ✗ | ✓ |
| Receding-horizon baseline | ✗ | ✗ | ✓ |
| Robustness vs a model-based controller | ✗ | ✗ | ✓ |
| Crossover vs that controller | ✗ | ✗ | ✓ |
| Monte Carlo (100 scenarios) | ✗ | ✗ | ✓ |
| Interactive workstation | ✗ | ✗ | ✓ |

Mo *et al.*: PPO over a mixed action space (tool change + parameters); the state carries $V_B$, breakage and $R_a$; objectives are quality and efficiency.
Wang *et al.*: MATD3 for batch milling; objectives are dimensional quality and machining time.
Key differentiator: **neither reference treats energy as an objective**. The speed-dependent $k_c$ creates a genuine energy/throughput trade-off, and one conditioned policy covers the whole frontier in a single run.

# Key Physics Equations (MillingDT/config/physics.py)

$$v_c = \frac{\pi D n}{1000} \tag{1}$$

$$\mathrm{MRR} = v_c \cdot a_p \cdot f_r \tag{2}$$

$$t_{\mathrm{cut}} = \frac{L_{\mathrm{cut}}}{n f_r} \tag{3}$$

$$\Delta V_B = C_{\mathrm{wear}} v_c^{\alpha} f_r^{\beta} a_p^{\gamma} t_{\mathrm{cut}} \tag{4}$$

$$R_a = k_1 f_r^2 + k_2 V_B + \frac{k_3}{v_c} \tag{5}$$

$$k_c(v_c) = k_{c0} \left( \frac{v_c}{v_{\mathrm{ref}}} \right)^m \tag{6}$$

$$P_{\mathrm{total}} = P_{\mathrm{idle}} + k_c(v_c) \cdot \mathrm{MRR} \tag{7}$$

$$E_{\mathrm{pass}} = P_{\mathrm{total}} \cdot t_{\mathrm{cut}} \tag{8}$$

## Key Insight: Interior Minimum

With $m = 0.6 > 0$:

- Idle energy $\propto 1/(n f_r) \downarrow$ with speed
- Cutting energy $\propto n^m \uparrow$ with speed
- Part energy $E_{\mathrm{part}}(v_c)$ has **interior minimum**
- Energy weight $w_2$ selects operating point

Wear: $C_{\mathrm{wear}} = 1.0 \times 10^{-2}$, anchored on a published tool life (about 9 parts per tool): none of the three reference papers carries a Taylor wear model.

# One Gymnasium Environment: `MillingDTEnv-v1`

**Two registrations**

- `env.ENV_ID` = `MillingDTEnv-v1`, 200-step limit
- `env.BATCH_ENV_ID` = `MillingDTBatchEnv-v1`
- The energy weight and the material are *inputs*, not separate tasks

**Observation (11-D, always the same shape)**

- 0–7: tanh-squashed $V_B$, $\kappa_{\text{rem}}$, previous $n, f_r, a_p$, $\mathrm{MRR}/\mathrm{MRR}_{\max}$, tanh-squashed $R_a$, tool-broken flag
- 8–10: the context: $w_2$, $C_{\text{wear}}$ over $[0.6, 1.4]$, $k_1$ over $[0.8, 1.2]$ of nominal

**Action**: $[-1, 1]^3 \to$ clipped $\to$ machine envelope

**Reward per step**

$$r = 100\left(w_1 \frac{\mathrm{MRR}}{\mathrm{MRR}_{\max}} - w_2 \frac{E_{\text{pass}}}{E_{\text{ref}}}\right) + 20\,\Delta\left(\frac{\kappa_{\text{removed}}}{\kappa_{\text{total}}}\right) \tag{9}$$

$$- 50\,\mathbb{1}[V_B > V_{B,\text{safe}}] - 100\,\mathbb{1}[\text{broken}] \tag{10}$$

$$- 50\,\mathbb{1}[\text{finish} \wedge R_a > R_{a,\text{safe}}] - 0.05 \tag{11}$$

- $+200$ when a part is finished in spec, plus a Pareto margin bonus of at most $+5$
- $w_1 = 1 - w_2$; safety limits $V_{B,\text{safe}} = 0.255$ mm and $R_{a,\text{safe}} = 1.36$ $\mu$m
- breakage $p = 0.3 \cdot$ wear factor $\cdot$ (load factor)$^2$ above $V_B > 0.15$ mm
- the scale is a design decision: the objective term is worth about 50 points per part, a quarter of a part

# Domain Randomization

- **Workpiece:** $\kappa_{\text{total}} \in [2, 15]$, $D \in [40, 150]$, $L_{\text{cut}} \in [50, 200]$
- **Material/Tool:** $C_{\text{wear}}$ at $[0.6, 1.4]\times$ nominal, $k_1, k_2, k_3$ at $[0.8, 1.2]\times$ nominal
- **Initial wear:** $V_{B,0} \sim \mathcal{U}[0, 0.8]\, V_{B,\max}$
- **Energy weight:** drawn $\mathcal{U}[0, 1]$ per episode under `-randomize-weight`
- **Shared tables:** same `MillingDT/config/scenarios.py` for training + Monte Carlo eval
- 
- **Measured:** the normalized physics features the policy sees on the training and the evaluation sets agree to 0.041 in the mean (std = 1), the same distribution
- **In the observation:** $C_{\text{wear}}$ and $k_1$ travel with the part, so the policy knows which material it is cutting
- **Belief and noise:** the context can carry a wrong material and the two measured features can carry noise, which is what the robustness experiment uses

# PPO Training Setup (Stable-Baselines3)

Table: Hyperparameters

| Parameter | Value |
|---|---|
| Policy | MlpPolicy (2×256) |
| Learning rate | $3 \times 10^{-4}$ (linear) |
| *n_steps* | 2048 |
| Batch size | 64 |
| *n_epochs* | 10 |
| $\gamma$ / GAE $\lambda$ | 0.99 / 0.95 |
| Clip range | 0.2 |
| Entropy coeff | 0.02 |
| VecNormalize | norm_obs=True, norm_reward=True |
| Envs / Device | 4 (`SubprocVecEnv`) / CPU |

## Training Runs

- **Production policy:** 1M steps, seed 42, `-randomize -randomize-weight` → `results/run_PPO_seed42/`
- **Ablation grid:** 7 $w_2$ × 3 seeds = 21 policies, 150k steps each
- **Batch-aware policy:** 1M steps on `MillingDTBatchEnv-v1`; **noisy-measurement policy:** 1M steps with `-obs-noise 0.05`

Every ablation policy has its weight pinned at training and evaluated at that same weight; the production policy sees the weight drawn per episode.

Heartbeat callback every 1000 steps → stall detection

# Production Policy: Monte Carlo Sweep (100 Scenarios)

| Metric | Mean | Std | 95% CI |
|---|---|---|---|
| MRR ($mm^3$/min) | 1608.5 | 550.7 | [1499.2, 1717.8] |
| Energy (kJ) | 66.4 | 48.0 | [56.9, 75.9] |
| Cycle time (s) | 11.4 | 6.9 | [10.1, 12.8] |
| Max $R_a$ ($\mu$m) | 1.610 | 0.178 | [1.575, 1.645] |
| Max $V_B$ (mm) | 0.1407 | 0.0672 | [0.1274, 0.1540] |

| Constraint | Rate | 95% CI |
|---|---|---|
| $R_a$ viol., worst pass | 55.0% | [45.2, 64.8]% |
| $R_a$ viol., finished surface | 4.0% | [0.2, 7.8]% |
| $R_a$ above safety limit, worst | 87.0% | [80.4, 93.6]% |
| $R_a$ above safety limit, finished | 4.0% | [0.2, 7.8]% |
| $V_B > 0.3$ mm | 0.0% | [0, 0]% |
| Tool breakage | 6.0% | [1.3, 10.7]% |
| Parts completed in spec | 94.0% | |

## Key Findings

- A much more aggressive operator than under the old reward: 1608.5 $mm^3$/min against 1205.2, and 11.4 s per part against 20.3 s
- The price is paid in the tool and in the part quality: 6.0% tool breakage and 4.0% of the finished surfaces outside the roughness limit, 87% of the worst passes above the safety margin
- Wear is not the binding constraint on a single part (0% violations, mean 0.1407 mm); it binds on the aggregate of a batch
- Mean operating point: 7907 RPM, 0.318 mm/r, 2.07 mm, 4.0 passes

# Energy Weight Ablation: Empirical Pareto Frontier

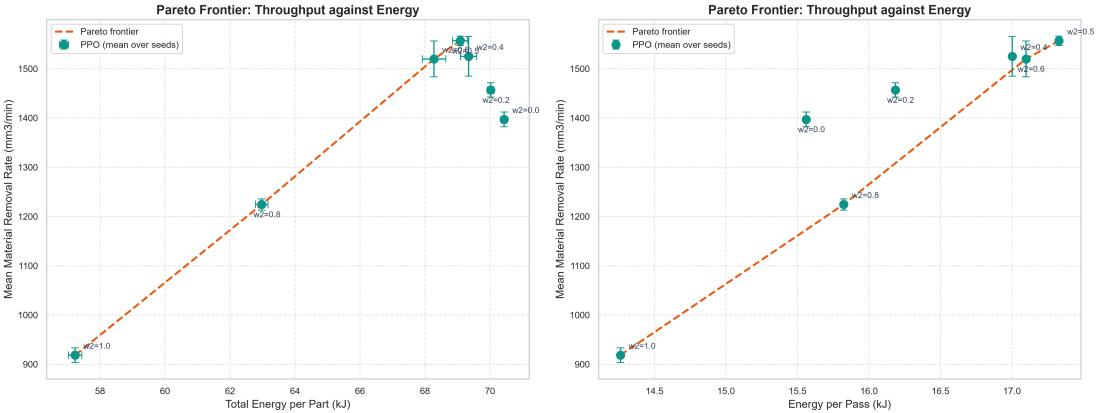
Table: Ablation: mean ± std over 3 seeds, 50 scenarios each, weight pinned

| $w_2$ | MRR | Energy (kJ) | Cycle (s) | $n$ (RPM) | $f_r$ | Frontier |
|---|---|---|---|---|---|---|
| 0.0 | 1397.1±14.8 | 70.4±0.1 | 14.8 | 7998 | 0.299 | dominated |
| 0.2 | 1456.9±14.6 | 70.0±0.1 | 13.8 | 7992 | 0.299 | dominated |
| 0.4 | 1524.9±40.2 | 69.3±0.2 | 13.0 | 7896 | 0.296 | dominated |
| 0.5 | 1556.6±9.7 | 69.1±0.2 | 12.7 | 7859 | 0.296 | ✓ |
| 0.6 | 1519.8±36.4 | 68.3±0.4 | 13.1 | 7638 | 0.296 | ✓ |
| 0.8 | 1224.4±11.3 | 63.0±0.2 | 15.4 | 6573 | 0.292 | ✓ |
| 1.0 | 918.5±14.9 | 57.2±0.2 | 20.1 | 5382 | 0.292 | ✓ |

## Frontier Analysis

From $w_2 = 0.5$ to $w_2 = 1.0$ the removal rate falls 41% (1557 → 919 mm$^3$/min) and the energy falls 17% (69.1 → 57.2 kJ). The trade-off is **asymmetric**: the energy is bought with a lot of removal rate, and with time (12.7 → 20.1 s). The spindle speed falls with the energy weight (7859 → 5382 RPM). Dominated points ($w_2 = 0.0, 0.2, 0.4$) remove less for more.

# Ablation Frontier: Removal Rate against Energy




Seven pinned-weight policies, three seeds each. The four non-dominated points are joined into the frontier: $w_2 = 0.5, 0.6, 0.8, 1.0$, and the three dominated points sit off it. Left panel: the energy of a part. Right panel: the energy of a pass, which the reward charges.

# One Policy Against 21 Policies

| $w_2$ | MRR, one policy | MRR, per weight | MRR gap | Energy, one policy | Energy, per weight | Energy gap |
|---|---|---|---|---|---|---|
| 0.0 | 1475.9 | 1397.1 | +5.6% | 68.4 | 70.4 | −2.9% |
| 0.2 | 1518.0 | 1456.9 | +4.2% | 65.8 | 70.0 | −6.1% |
| 0.4 | 1608.5 | 1524.9 | +5.5% | 66.4 | 69.3 | −4.2% |
| 0.5 | 1630.3 | 1556.6 | +4.7% | 66.0 | 69.1 | −4.4% |
| 0.6 | 1540.8 | 1519.8 | +1.4% | 65.5 | 68.3 | −4.0% |
| 0.8 | 1114.5 | 1224.4 | −9.0% | 57.9 | 63.0 | −8.0% |
| 1.0 | 720.4 | 918.5 | −21.6% | 51.7 | 57.2 | −9.7% |

## Result (the claim of the previous revision has reversed)

Over the weights from 0.0 to 0.6 the single policy removes 1.4% to 5.6% more material for 2.9% to 6.1% less energy. At $w_2 = 0.8$ and $w_2 = 1.0$ it is behind by 9.0% and 21.6% of the removal rate. The weight in the observation is a weaker signal than a dedicated policy where the trade-off is steepest.

The same policy swept at runtime: MRR 1475.9 ($w_2 = 0.0$) → 720.4 ($w_2 = 1.0$), energy 68.4 → 51.7 kJ, spindle 8000 → 4111 RPM. **The weight stays a runtime query**, not a retrain, but it is not free at the ends of the range.

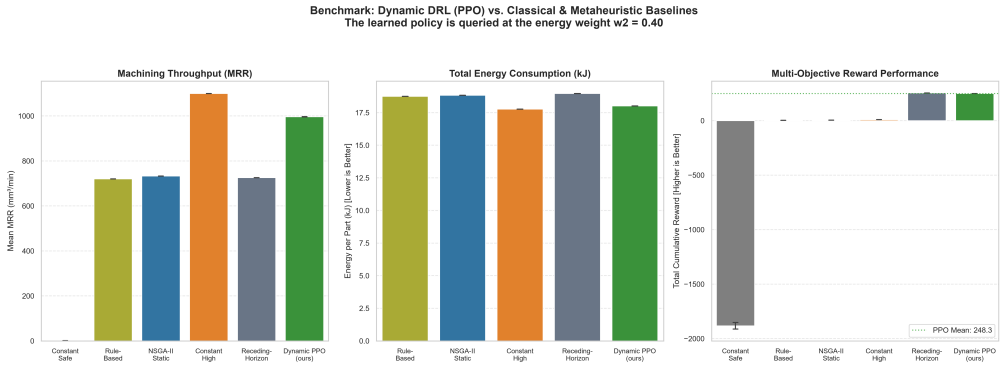
# Baseline Comparison (50 Episodes, Nominal Condition, $w_2 = 0.4$)

Table: Baseline comparison on the same 50 episodes

| Policy | Reward | MRR | Energy (kJ) | Cycle (s) | $R_a$ final ($\mu$m) |
|---|---|---|---|---|---|
| Constant high | 10.0 | **1099.6** | 17.8 | **4.3** | 1.56 (violates) |
| NSGA-II static | 5.1 | 732.9 | 18.8 | 6.4 | 1.56 (violates) |
| Rule based | 3.3 | 719.8 | 18.8 | 6.6 | 1.56 (violates) |
| **Receding-horizon** | 253.5 | 725.5 | 19.0 | 6.7 | 1.25 |
| Constant safe | -1882.9 | 1.6 | 385.7 | 768.0 | 0.26 |

- **Same 50 episodes** of the nominal condition for all six policies, at $w_2 = 0.4$
- **Three of the four static policies finish outside the roughness safety limit** (1.56 $\mu$m against 1.36 $\mu$m): no completion bonus, the finishing penalty, and they land near zero
- Constant-Safe stays inside the limit (0.26 $\mu$m) but needs 768 s and 385.7 kJ per part: $-1882.9$
- **Receding-horizon** is the strongest baseline, $+253.5$, and the only classical policy that finishes every part in spec; it pays with throughput: 725.5 mm$^3$/min and 6.7 s

# Policy Comparison: Six Policies, Same 50 Episodes



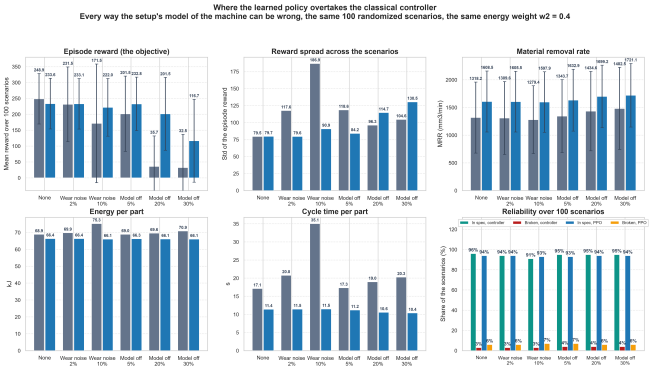

The three fast static policies land near zero: they finish outside the roughness safety limit, so they are paid no completion bonus and are charged the finishing penalty. Constant-Safe stays inside but needs 768 s per part. The comparison that matters is the receding-horizon controller against the learned policy, the only two with a positive reward.

# Robustness Against the Classical Controller

| Regime | Arm | Reward | Std | MRR | kJ | s | Broken | Done |
|---|---|---|---|---|---|---|---|---|
| Controller has every advantage | Receding-horizon | 248.9 | 79.5 | 1318.2 | 68.9 | 17.1 | 3% | 96% |
| | PPO | 233.6 | 79.7 | 1608.5 | 66.4 | 11.4 | 6% | 94% |
| Wear measurement is noisy | Receding-horizon | 181.3 | 185.9 | 1300.4 | 74.2 | 31.1 | 4% | 91% |
| | PPO | 226.8 | 87.2 | 1602.1 | 66.3 | 11.5 | 7% | 93% |
| | PPO trained with noise | 213.6 | 87.3 | 1554.2 | 66.9 | 12.0 | 6% | 94% |

The same 100 randomized scenarios at $w_2 = 0.4$ in every row. The regime changes what the controller is given, not what the policy is given: the exact model, the exact state and the material in its hand in the first regime, a 5% noise on the wear and the remaining stock in the second. Fairness: both arms are the general ones, neither is trained for the noise, and the noise is tilted against the policy, which receives it on both measured quantities while the controller receives it on the wear alone. The figure on the next slide puts all six ways the setup's model of the machine can be wrong side by side.

# Robustness: Six Ways the Setup's Model Can Be Wrong




Six panels: the reward, its spread, the removal rate, the energy, the cycle time and the reliability. The six cases are the perfect twin, the wear measurement noisy by 2% and by 10%, and the material model off by 5%, 20% and 30%. The values are in the tables. Level on the reward in the perfect twin (248.9 against 233.6) and ahead on the throughput, the energy and the cycle time. At 10% of noise the controller falls to 171.5 and the policy holds at 222.0; with the material model off by 20% the controller falls to 35.7 and the policy holds at 201.5.

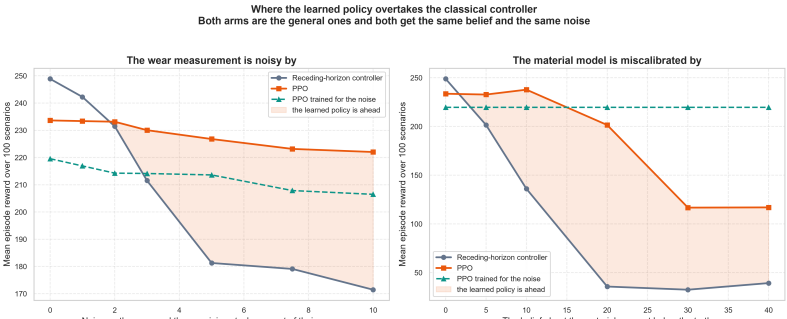
# Robustness: How to Read It

1. **Inside a perfect digital twin the two methods are comparable, and the measured gap is partly an artifact of the reward.** The controller computes the optimum of the same objective at every pass with the exact model and the exact state, and it scores 248.9 against the policy's 233.6. The gap is inside the noise: the standard error of the mean is about 8 for both arms, so a difference of 15 has a $t$ of about 1.4. And about nine of those fifteen points are the reward: the controller cuts the part in 5.85 passes against the policy's 4.00 and is paid for every pass.
2. **The learned policy is the better throughput operator.** It removes 22% more material per minute (1608 against 1318), finishes a part in a third less time (11.4 against 17.1 s) and spends 3.7% less energy. It pays with twice the tool breakage (6% against 3%).
3. **The advantage appears where a plant actually is.** With a 5% noise on the wear and on the remaining stock, the controller's reward falls from 248.9 to 181.3 and its spread doubles to 185.9 against the policy's 87.2. The policy trained with the same noise keeps 213.6 and cuts the part in 12.0 s against the controller's 31.1 s. The general policy scores 226.8 at that same 5% noise.

## The claim

Inside a perfect digital twin the two methods are comparable: the policy is ahead on the physical metrics and behind on the breakage, and the gap the reward does show is partly the artifact of how the reward pays for a rate. The advantage appears where the controller's information advantage disappears, and the sweep of the regime map puts a number on it: the learned policy is the better operator once the wear measurement is noisier than about 2% of its range.

# The Regime Map: Two Sweeps, One Headline




Mean episode reward over the same 100 randomized scenarios at $w_2 = 0.4$. Left: the measurement noise on the flank wear and the remaining stock, as a percent of their range. Right: the error in the setup's belief about the material, as a percentage below the truth. The shaded band is where the learned policy is ahead of the controller. The crossover is at about 2% of the range of the wear measurement: below it the classical controller is the better operator, above it the learned policy is. The noise sweep is the headline, because the noise is a property of the instrumentation and it applies to both arms in the same way.

# Regime Map: How Noisy the Wear Measurement Has to Be

| Measurement noise | Receding-horizon | PPO | PPO trained for the noise | PPO ahead by |
|---|---|---|---|---|
| 0.0% | 248.9 | 233.6 | 219.6 | −15.3 |
| 1.0% | 242.3 | 233.4 | 216.9 | −8.9 |
| 2.0% | 231.5 | 233.1 | 214.3 | +1.7 |
| 3.0% | 211.5 | 230.0 | 214.1 | +18.5 |
| 5.0% | 181.3 | 226.8 | 213.6 | +45.5 |
| 7.5% | 179.1 | 223.2 | 207.9 | +44.1 |
| 10.0% | 171.5 | 222.0 | 206.5 | +50.6 |

Mean episode reward over the same 100 randomized scenarios at $w_2 = 0.4$. The noise is a fraction of the range of the flank wear and of the remaining stock. The last column is the general policy's reward minus the controller's. This sweep is the headline: the noise is the fair test, a property of the instrumentation that applies to both arms alike.

1. **The crossover is at about 2% of the range of the wear measurement.** Below it the classical controller is the better operator, above it the learned policy is.
2. **The two degrade differently.** The controller falls from 248.9 to 171.5, a 31% loss, because a noisy reading makes its plan violate the limits and break tools. The policy falls from 233.6 to 222.0, a 5% loss. A model-based controller has a cliff, a learned policy has a slope.
3. **Training for the noise does not pay.** The noise-trained policy is below the general policy at every noise level, including the 5% it was trained for, 213.6 against 226.8. Its robustness comes from training across a distribution of scenarios.

# Regime Map: How Wrong the Material Model Can Be

| Material model off by | Receding-horizon | PPO | PPO trained for the noise | PPO ahead by |
|---|---|---|---|---|
| 0% | 248.9 | 233.6 | 219.6 | −15.3 |
| 5% | 201.5 | 232.8 | 219.6 | +31.2 |
| 10% | 136.0 | 237.8 | 219.6 | +101.8 |
| 20% | 35.7 | 201.5 | 219.6 | +165.8 |
| 30% | 32.5 | 116.7 | 219.6 | +84.3 |
| 40% | 39.3 | 117.0 | 219.6 | +77.7 |

Mean episode reward over the same 100 randomized scenarios at $w_2 = 0.4$. The belief about the material is a percentage below the truth, so the setup under-rates the wear and the roughness of the material. The same belief goes to both arms; the measurement noise is zero.

1. **The controller's tolerance of a wrong material model is about 2% to 3%.** A 5% error costs it 47 points (248.9 to 201.5) and a 20% error costs it 213 (to 35.7): it solves with the belief it is given, so a belief that under-rates the wear produces a plan that violates the limit.
2. **The learned policy holds to a 20% error and collapses at 30%.** At 20% it scores 201.5, 13% below its own no-error score of 233.6, and at 30% it falls to 116.7, because the belief has left the distribution it was trained on and its context feature is clipped at the edge of the band. That limit is a design parameter: widen the randomization band and the tolerance widens with it.
3. **A conservative policy is insensitive to a wrong model.** The policy trained for the noise is flat at 219.6 across the whole range, because it keeps the tool away from its limit whatever it believes about the material. It pays for that with the reward: it is below the general policy everywhere.

# The Regime Map: The Fairness Decisions

## The fairness decisions

- **Noise sweep.** Both arms of the headline comparison are the general ones, neither is trained for the noise. The noise is a property of the instrumentation and not of the method: it changes only what the instrument reports, and it applies to both arms in the same way. The noise is tilted against the policy: the controller gets a noisy reading of the flank wear alone, the policy gets the noise on the flank wear and on the remaining stock. The noise-trained policy is a third, clearly labelled arm, and it is not the headline, because a controller can also be prepared for a noisy instrument by widening its safety margin, and that is a design choice rather than a training one.
- **Model error sweep.** The same belief about the material goes to both arms: the controller solves with it and the policy reads it in its context. The error is in the damaging direction, because a setup that under-rates the wear and the roughness of the material it is cutting will push the tool harder than the material can take. Both arms are the general ones, neither is prepared for the error, and the noise model is tilted against the policy wherever noise is present: the policy receives it on both measured quantities while the controller receives it on the wear alone.

## The claim

The learned policy is the better operator once the wear measurement is noisier than about 2% of its range, and the classical controller is the better operator below that, inside a perfect digital twin. The model error sweep supports that boundary and does not carry it.

# Reward Sensitivity: How the Reward Pays for a Rate

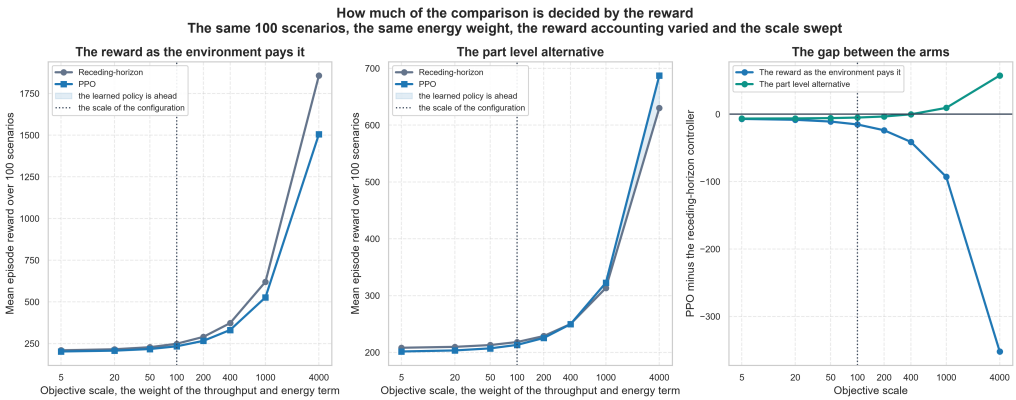
The reward pays the removal rate of every pass, the standard way to express a rate in a reward that is paid per pass. The total credit therefore grows with the number of passes a plan takes, so a plan that cuts the same part in more, shallower passes is paid more for it.

On the same 100 randomized scenarios at $w_2 = 0.4$, measured by `experiments/reward_sensitivity.py`: the controller takes 5.85 passes against the policy's 4.00 and collects 27% more objective credit (+0.413 against +0.326) for the same material.

| | Shipped form (sum of the per-pass rates) | | | Alternative form (material against energy) | | |
|---|---|---|---|---|---|---|
| **Objective scale** | **Controller** | **Policy** | **Gap** | **Controller** | **Policy** | **Gap** |
| 5 | 209.7 | 202.6 | +7.1 | 208.2 | 201.6 | +6.5 |
| 20 | 215.9 | 207.5 | +8.3 | 209.7 | 203.4 | +6.3 |
| 50 | 228.3 | 217.3 | +10.9 | 212.9 | 207.1 | +5.8 |
| 100 | 248.9 | 233.6 | +15.3 | 218.2 | 213.2 | +5.0 |
| 200 | 290.1 | 266.2 | +23.9 | 228.7 | 225.3 | +3.4 |
| 400 | 372.7 | 331.5 | +41.2 | 249.9 | 249.6 | +0.2 |
| 1000 | 620.2 | 527.2 | +93.0 | 313.2 | 322.6 | −9.4 |
| 4000 | 1857.9 | 1506.0 | +352.0 | 630.1 | 687.4 | −57.4 |

Mean reward of each arm on the same 100 scenarios, under two objective forms and eight objective scales. The gap is the controller's mean reward minus the policy's.

# Reward Sensitivity: How Much of the Comparison the Reward Decides




Three panels: the reward as the environment pays it, the part level alternative, and the gap between the arms. The scale is not the lever: the controller leads at every scale and the gap grows with it, 7.1 points at 5 and 352.0 at 4000. The accounting of a rate is: under the part level alternative the two close to 5.0 points at the scale of 100, and the policy overtakes by 9.4 points at 1000 and 57.4 at 4000.

# Reward Sensitivity: Three Checks and the Consequence

1. **The objective scale is not the lever**: the controller leads at every scale from 5 to 4000 and the gap only grows with it, 7.1 points at 5 and 352.0 at 4000, so the sign never flips.
2. **The part level alternative was implemented and rejected**: it removes the pass count from the accounting, so the arms are much closer (5.0 points at the scale of 100, against 15.3) and the policy is ahead at the large scales (9.4 points at 1000, 57.4 at 4000). But the material a part gives up is fixed by its stock, so the throughput weight would be inert and the weight sweep would stop measuring what it claims to measure.
3. **What a correct form would need**: a rate cannot be accumulated over a part without becoming a volume, so a reward has to either price the cycle time directly or compare the part's average rate to a reference at the end of the part. Both change the meaning of the weights, and the second makes the objective sparse and the credit assignment poor.

## The consequence

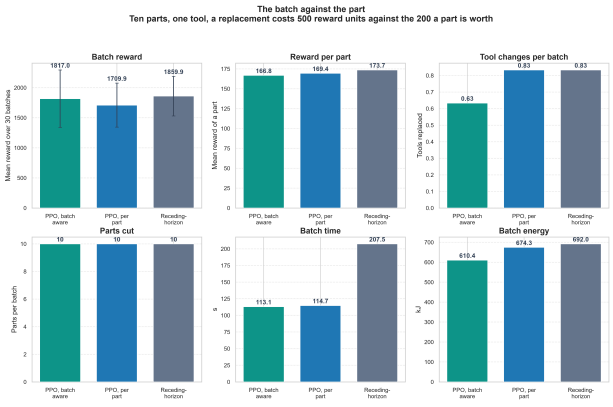
The ideal case reads as a tie, not a win: the policy is ahead on the physical metrics (22% more material removed, a third less cycle time, 3.7% less energy) and behind on the tool breakage (6% against 3%), and the gap the reward does show is partly the reward's artifact. The reward as shipped carries it, documented rather than hidden.

# Batch Against the Part

| Arm | Batch reward | Std | Reward per part | Tool changes | Batch time (s) | Batch energy (kJ) |
|---|---|---|---|---|---|---|
| PPO, batch aware | 1817.0 | 477.8 | 166.8 | 0.63 | 113.1 | 610.4 |
| PPO, per part | 1709.9 | 366.1 | 169.4 | 0.83 | 114.7 | 674.3 |
| Receding-horizon, per part | 1859.9 | 328.8 | 173.7 | 0.83 | 207.5 | 692.0 |

- A batch is ten parts cut with one tool; the tool is replaced when it breaks or reaches its wear limit, and the replacement costs 500 reward units against the 200 a finished part is worth. The observation carries the tool life left and the fraction of the batch left.
- The batch context is worth 24% fewer tool changes and 9.5% less energy at the same completion: the policy that sees the tool life and the batch that is left replaces the tool 0.63 times per batch against 0.83, and uses 610.4 kJ against 674.3 kJ. It also collects more batch reward, 1817.0 against 1709.9.
- The per-part controller gets the highest batch reward, 1859.9, and buys it with the machine time: 207.5 s for the batch against the batch-aware policy's 113.1 s, about 1.8 times the machine time for a batch reward that is 2.4% higher. It solves one part at a time, so it cannot price a tool change against the parts that follow it.

# Batch Against the Part: One Tool, Ten Parts




Six panels: the batch reward, the reward per part, the tool changes, the parts cut, the batch time and the batch energy. The values are in the table. The batch-aware policy replaces the tool 0.63 times per batch against 0.83, spends 610.4 kJ against 674.3 and 692.0, and cuts the batch in 113.1 s against 207.5 s for the per-part controller.

# Controller Initialization: Method and Result

**Question**: does a policy that starts from the controller end up better than the expert it started from?

## Method

Record the controller on 2000 randomized episodes → fit a supervised actor (2×256, Tanh) on those trajectories by regression → copy the fitted weights into a fresh PPO policy → train with the production budget (1M steps) and evaluate on 100 scenarios at $w_2 = 0.4$. `MillingDT/agent/controller_init.py` `experiments/controller_experiment.py`

| Metric | Receding-horizon controller | PPO from scratch | PPO initialized from controller |
|---|---|---|---|
| Mean episode reward | 248.9 | 233.6 | 119.8 |
| Reward std | 79.5 | 79.7 | 178.7 |
| MRR ($mm^3$/min) | 1318.2 | 1608.5 | 1274.9 |
| Energy per part (kJ) | 68.9 | 66.4 | 81.5 |
| Cycle time (s) | 17.1 | 11.4 | 42.6 |
| $R_a$ viol., finished | 2% | 4% | 4% |
| Flank wear violations | 1% | 0% | 9% |
| Tool breakage | 3% | 6% | 4% |
| Parts completed | 96% | 94% | 87% |

# Controller Initialization: What It Shows

## The initialization is now harmful

- The initialized policy scores 119.8 against the from-scratch policy's 233.6, a gap of 113.8, more than fourteen times the standard error of the mean
- The clone starts the policy at the controller's conservative operating point, and with the wear constraint binding that conservatism is expensive
- It cuts a part in 42.6 s against 17.1 s and 11.4 s, and spends 81.5 kJ against 66.4 kJ
- It is the only arm that exceeds the flank wear limit (9%)

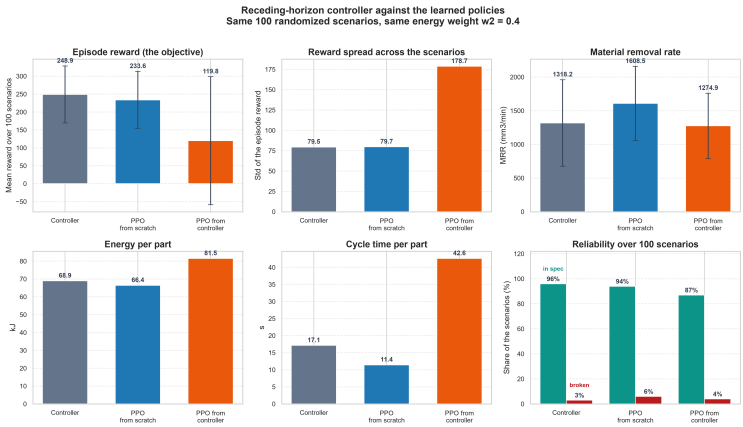
## The result reversed

- The earlier revision found the initialization harmless: it made the policy more consistent and did not change the reward
- That was the reward not counting the cost of the slow operating point
- With the throughput counted, the initialization is what the measurement says it is
- What stays is the record and clone machinery, not the initialization itself

## What the work claims

The policy and the model-based controller are comparable inside a perfect digital twin, and the gap the reward shows is partly its artifact. The policy is the better operator once the wear measurement is noisier than about 2% of its range, and the classical controller is the better operator below that.

# Controller Initialization: The Three Arms




Six panels, too detailed to read at slide size: the values are in the table. The initialized arm sits 114 points below the from-scratch arm on the reward, and its spread of 178.7 is more than twice the spread of the other two: the initialization does not buy consistency any more, it buys variance.

# Transfer Experiment: Cross-Material Adaptation

**Target material (harder alloy, same machine):** $k_1 \times 1.25$, $k_2 \times 1.30$, $C_{\text{wear}} \times 2.0$. The source policy was trained for 300,000 steps on the nominal material; the adaptation budget on the target is 50,000 steps.

| **Arm** | **Reward** | $R_a$ **viol.** worst pass | $R_a$ **viol.** finished surface | **Parts** completed |
|---|---|---|---|---|
| Source policy, zero shot | 168.5 | 100.0% | 6.7% | 86.7% |
| Source policy, fine tuned (50k) | 101.9 | 26.7% | 0.0% | 86.7% |
| Trained on the target from scratch (50k) | 101.9 | 26.7% | 0.0% | 86.7% |
| **Production policy, zero shot** | 190.1 | 100.0% | 10.0% | 90.0% |

## How to read this

The production policy was trained across a **distribution of materials** with the material context in its observation, so the harder material is inside its training distribution and it needs no adaptation at all: it scores the highest reward (190.1) and completes 90.0% of the parts. Its worst pass violates the roughness limit in every scenario, and its finished surface in 10.0% of them.

Adaptation repairs the roughness and costs more than it gains: fine tuning the nominal-trained policy for 50,000 steps, or training on the target from scratch, both cut the worst-pass violations to a quarter and land at 101.9 against the source policy's 168.5 zero shot. The target alloy wears the tool twice as fast, and a policy that slows down for that trades away more throughput than it gains in specification.

# Streamlit Digital Twin Workstation

- **2D Stock Depletion:** Cylindrical billet cross-section; pass-by-pass cutout; remaining $\kappa$, $a_p$; tool engagement
- **Micro-Topography:** Feed-mark cusp $y_{\text{cusp}}(x) \approx x^2/(8r_\epsilon) + \text{noise}$; ISO 1302 $R_a$ overlay; 1.6 $\mu$m ceiling
- **G-Code Generator:** ISO 6983 Fanuc/Siemens compliant; exact PPO sequence; tool/diameter aware
- **Multi-Controller:** Side-by-side PPO vs NSGA-II vs Rule-Based vs Constant-High; live telemetry
- **Cached Loading:** `@st.cache_resource` loads model once per session

```
streamlit run app.py
```

# Architecture & Reproducibility

**Module Structure**

- `MillingDT/`: config, env (part + batch), agent (controller init), baselines, sim, utils
- `experiments/`: train, eval, ablation, baselines, transfer, controller, robustness, regime map, batch, reward sensitivity, pipeline
- `viz/`: shared plotting utilities
- `results/`: one directory per run

**Reproducibility**

- Global seeding (Python, NumPy, PyTorch, CUDA)
- Shared scenario tables (train = eval)
- `experiments/summarize_results.py` regenerates `RESULTS.md`
- No unused imports (AST verified)

**Test Coverage (44 tests)**

- 23 env tests: clipping, truncation, soft norm, 11-D observation, constants, overrides, energy norm, Gym checker, randomization, scenario scales, context (weight, material), observation noise, energy minimum, reward golden values
- 11 batch env tests: the 13-D observation, the tool life and the batch left, the wear carried over, the tool change, the batch reward, the seeded reproducibility
- 5 integration, 3 callback, 2 training: roundtrip, registration, context, heartbeat, convergence

# Discussion: What Worked

## What Worked

- Speed-dependent $k_c$ creates a genuine trade-off
- The scale of the reward decides the operating point: 50 points per part against 200 for a part
- Domain randomization $\rightarrow$ 94% completion
- One policy covers the range, ahead over $w_2 = 0.0$ to 0.6
- Training across materials makes the adaptation unnecessary

## What Worked (continued)

- A controller that prices the whole reward is a fair, strong baseline (184.3 $\rightarrow$ 248.9)
- Robustness: comparable inside a perfect twin; the advantage appears where the controller's information runs out
- The regime map: the policy is the better operator once the wear measurement is noisier than about 2% of its range
- A second sweep, a wrong material model, supports it: the controller tolerates 2% to 3% of that error, the policy 20%
- Measuring the reward found the artifact a fair comparison would have hidden
- Batch context: 24% fewer tool changes, 9.5% less energy

# Discussion: Limitations and Future Work

## Limitations

- Inside a perfect twin the two are comparable: 248.9 against 233.6, the gap inside the noise
- The learned policy is the better throughput operator, at twice the tool breakage
- The crossover at 2% is a property of this controller, this reward and this noise model
- The tolerance to a wrong material model is set by the training band: 20% holds, 30% collapses
- The comparison is one noise model deep: one degraded regime, and the regime map sweeps that noise and, separately, the material model error
- Initializing from the controller is now harmful: 119.8 against 233.6
- The reward pays a rate per pass, so the credit grows with the pass count
- The wear tool life varies by only a factor of a few across the envelope
- Ablation: 150k steps per weight, so the frontier is a lower bound; sim-only, no real calibration yet

## Future Work

- A reward that pays a rate correctly: price the cycle time directly, or compare the part's average rate to a reference
- Publish the sensitivity of any comparison to the reward form, as it is here
- Compare against a restricted controller (is the gap an information gap?)
- Measure the crossover under a controller prepared for a noisy instrument
- Measure the crossover on the noise of a real wear sensor, not on a Gaussian model of it
- A wider randomization band: retrain, repeat the model error sweep, and measure the error in the other direction
- Sim-to-real calibration, true multi-objective RL, online $w_2$ adaptation, batch-level planning

# Conclusion

- **Digital Twin** with speed-dependent cutting energy, tool wear, roughness, breakage; the wear coefficient is anchored on a published tool life
- **Two Gymnasium Environments** (the part, 11-D, the batch, 13-D) and **One Production Policy** (1M steps): the energy weight is a runtime query
- **Rigorous Evaluation**: Monte Carlo (100 scenarios, 95% CIs), ablation frontier, weight sweep against the 21 policies
- **Baselines**: five classical policies on the same 50 episodes; the receding-horizon controller is the strongest at +253.5
- **Robustness**: comparable inside a perfect twin (248.9 against 233.6, the gap inside the noise); with a noisy wear measurement the controller falls to 181.3, the policy holds at 226.8
- **Regime map**: two sweeps. The noise sweep carries the claim: the policy is the better operator once the wear measurement is noisier than about 2% of its range. The model error sweep supports it: the controller tolerates 2% to 3% of a wrong material model, the learned policy 20%
- **Reward sensitivity**: the credit grows with the number of passes (27% more for the controller)
- **Batch**: 24% fewer tool changes and 9.5% less energy; 207.5 s against 113.1 s for the batch
- **Controller Initialization**: the result reversed, the initialization is now harmful (119.8 against 233.6)
- **Transfer Experiment**: the production policy needs no adaptation; the nominal-trained one does
- **Interactive Workstation** and **full reproducibility**: every result table from the committed artifacts by one script

## Key Scientific Finding

The speed-dependent specific cutting energy creates a **genuine energy/throughput trade-off**, and DRL learns to navigate it pass-by-pass: across the frontier the removal rate falls 41% while the energy of the part falls 17%. Against a model-based controller the claim is a boundary, not an advantage: inside a perfect digital twin the two are comparable, and the learned policy is the better operator once the wear measurement is noisier than about 2% of its range.

# Key Commands

| Task | Command |
|---|---|
| Tests | `pytest tests/ -v` |
| Production | `experiments.train_rl -steps 1000000 -randomize -randomize-weight` |
| Pinned weight | `experiments.train_rl -steps 150000 -randomize -w2 0.6` |
| Noisy measurement | `experiments.train_rl -steps 1000000 -obs-noise 0.05` |
| Batch policy | `experiments.train_rl -env-id MillingDTBatchEnv-v1 -steps 1000000` |
| Ablation grid | `experiments.train_rl_ablation -steps 150000 -slots 7` |
| MC eval | `experiments.evaluate -scenarios 100` |
| Weight sweep | `experiments.evaluate -sweep-weight -scenarios 100` |
| Compare | `experiments.evaluate -compare-only` |
| Baselines | `experiments.run_baselines` |
| Transfer | `experiments.transfer_experiment 300000 50000` |
| Controller init | `experiments.controller_experiment 1000000 2000` |
| Robustness | `experiments.robustness_experiment` |
| Regime map | `experiments.regime_map` |
| Batch | `experiments.batch_experiment` |
| Reward sensitivity | `experiments.reward_sensitivity` |
| Results | `experiments.summarize_results` |
| Workstation | `streamlit run app.py` |
| Full pipeline | `experiments.run_pipeline -steps 1000000 -ablation_steps 150000` |

Each entry is run as `python -m <module>` from the project root.

# Thank You

Questions & Discussion

Repository: `G:/My Drive/IITM/AI in Mfg`
ME6324 – AI in Manufacturing | IIT Madras